QP CODE: 25900864

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
FIRST SEMESTER MGU-UGP (HONOURS) REGULAR
EXAMINATION, NOVEMBER 2025 First Semester
Discipline Specific Core Course - MG1DSCHTC100 - BASIC HORTICULTURE
(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A. Short Answers
Answer any ten questions. Each question carries 1 mark.

1. **What is the role of soil preparation in the successful establishment of a garden?**
   - Improves soil structure and aeration; enhances drainage; facilitates root penetration; removes weeds and pests; ensures uniform nutrient availability.

   (Give marks for any two relevant point related to soil health and plant nutrition/growth)

2. **What are the key features of a tropical garden?**
   - Dense evergreen plants; bright flowering species; lush foliage; high humidity; layered planting; shade-loving ornamentals. (**Any two relevant points**)
3. **What is inarching?**
   - A grafting method where a scion still attached to the parent plant is joined to the rootstock until union occurs.**(Liberal valuation)**
4. **What is water potential?**
   - The potential energy of water in a system; determines direction of water movement from higher to lower potential.
5. **What is meant by a bud?**
   - An undeveloped shoot/ initial, mass of undifferentiated cells containing a growing point; may develop into a leaf, flower, or branch.
6. **What is a vegetable zone?**
   - A geographic region suited for commercial vegetable cultivation based on climate, soil, and seasonal factors.
7. **Write a short note on Northeastern subtropical fruit zone.**
   - Includes Assam, Meghalaya, etc.; humid subtropical climate; supports citrus, pineapple, banana, papaya, guava.
8. **How does Horticulture improve human health and nutrition?**
   - Provides vitamins, minerals, antioxidants; promotes balanced diet; supports mental well-being; ensures nutritional security.
9. **Discuss the role of pH management in aquaponics.**
   - Maintains balance for fish, plants, and microbes; optimal nutrient availability; supports nitrification; prevents toxicity.**(Liberal valuation. Give marks if nutrient availability or any other relevant point is mentioned)**
10. **What is the function of a trencher in digging?**

- Cuts narrow, deep trenches for planting, irrigation lines or drainage; reduces labour and time.

11. **What is seed?**

- A fertilized mature ovule containing embryo, endosperm and seed coat; unit of reproduction.

12. **What is nutrient management?**

- Supplying plants with balanced nutrients through fertilizers, organic matter, and soil amendments to optimize growth, avoiding their wastage and leaching.
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Part B:
Short Essay Type Questions
Answer any six questions. Each question carries 5 marks.

13. **What are implements? Explain different types of implements used for land preparation.**
    - Implements are tools used in various agricultural operations (**Definition- 2 marks**)
    - Types of implements: Ploughs (mouldboard, disc), cultivators — break soil, invert clods.
    - Harrows, rollers — pulverize and level soil.
    - Hoes, weeders — weed control, soil aeration.
    - **Other implements:** Levellers, spades, rotavators or any tools used in land preparation.**(Types- 3 mark)**
14. **Explain the role of heating systems in hotbeds &how they benefit early plant growth.**
    - Role: Protection against low temperature and associated problems (**3 marks**).
    - **Benefits:** Early germination; prevents frost injury; enhances seedling vigour; extends growing season; supports nursery raising in winter (**2 mark**).
15. **Discuss the limitations of aeroponics in terms of crop selection.**
    - High initial cost; requires uninterrupted power; sensitive to system failures; nutrient clogging issues; limited crop range; requires technical expertise.**(Liberal Valuation)**
16. **What are the objectives of horticultural breeding programs in terms of plant traits like yield and taste?**
    - Higher yield; improved taste and quality; pest and disease resistance; stress tolerance; uniformity; shelf-life improvement; better colour and appearance.**(Liberal valuation)**
17. **How are drones used in horticulture operations?**
    - Aerial mapping; crop monitoring; pest and disease detection; precision spraying; fertilizer application; estimating canopy/plant health via sensors.
18. **Compare and contrast Surface irrigation and Sub-soil irrigation.**
    - **Surface irrigation:** Water applied over field surface; low cost; higher evaporation loss; uneven distribution.
    - **Sub-soil irrigation:** Water supplied underground via porous pipes; efficient water use; costly installation; suitable for orchards and greenhouses.
19. **What is the importance of ornamental gardening in institutions?**
    - Improves aesthetics; provides clean air; creates learning environment; promotes mental well-being; supports biodiversity; enhances institutional image.
20. **Explain the living components of a garden with their key functions.**
    - **Trees:** Shade, structure, microclimate.
    - **Shrubs:** Borders, screens.
    - **Herbs & groundcovers:** Cover soil, reduce erosion.

- **Climbers:** Vertical decoration.
- **Lawns:** Open space, recreation.
- **Flowering plants:** Colour, seasonal beauty.

Part C: Essay Type Questions
Answer any one question. Question carries 10 marks.

21. **Write an essay on plant growing structures.**
    - **Types:** Greenhouse, shade-net house, polyhouse, hot beds
    - **Components:** Frame, cladding material, ventilation, cooling/heating systems.
    - **Functions:** Controlled environment for temperature, humidity, light; off-season cultivation; seedling production; protected cultivation of flowers/vegetables.
    - **Advantages:** Higher productivity, quality produce, reduced pest attack.**(Or any other plant growing structure with relevant details)**
22. **Write an essay on different agriculture parameters monitored by artificial intelligence**
    - Soil moisture, fertility, pH; nutrient status; crop growth stage; disease/pest prediction; weather forecasting; irrigation scheduling; yield prediction; drone-based imaging; smart farm automation.
    - AI improves precision, reduces cost, increases sustainability.
      (Liberal valuation. Give full marks if role of AI in agriculture is mentioned)